Yacht Hydrodynamic Resistance

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Outline

Part 1

Data and Problem

The Yacht Hydrodynamics dataset

Source. UCI ML Repository — Delft Systematic Yacht Hull Series, towing-tank experiments.

- 22 hull forms, each tested at several speeds
- $n = 308$ observations (one hull $\times$ one speed per row)
- **Response:** residuary resistance per unit weight of displacement
- **Predictors:** speed (Froude number) $+$ five hull form factors, constant within a hull

Variables

`LCB`	centre of buoyancy (%)
`prismatic`	prismatic coefficient
`L_disp`	length–displ. ratio
`beam_draught`	beam–draught ratio
`L_beam`	length–beam ratio
`Froude`	$Fn = v/\sqrt{gL}$
`resistance`	response (target)

The five form factors vary only across the 22 hulls; $Fn \in [0.125, 0.45]$ is the dominant predictor.

Part 2

Log–log Regression

Model and reference prior

Goal. Verify the physical power law $\text{resistance} \propto Fn^{\gamma_F}$ with $\gamma_F \approx 4$. Taking logs makes the exponent a regression slope:

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(Fn) + \sum_{k=1}^{5} \beta_k \, \text{hull}_k + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

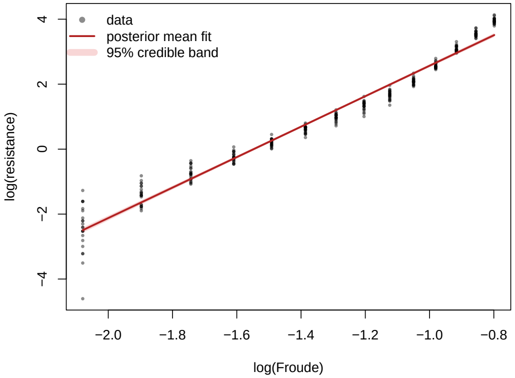
Baseline non-informative **Jeffreys reference prior**

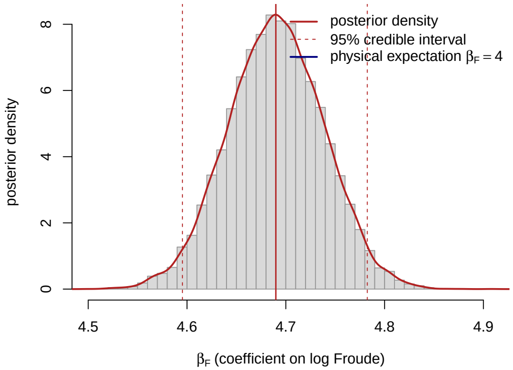
$$\pi(\boldsymbol{\beta} \mid \sigma^2) \propto 1, \qquad \pi(\sigma^2) \propto \sigma^{-2}.$$

Closed-form posterior

$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}(\hat{\boldsymbol{\beta}}, \, \sigma^2(\boldsymbol{X}^\top \boldsymbol{X})^{-1})$; marginalising σ^2 gives a multivariate Student-t with $n - p$ d.o.f. The 95% credible intervals coincide with the OLS confidence intervals — no simulation required.

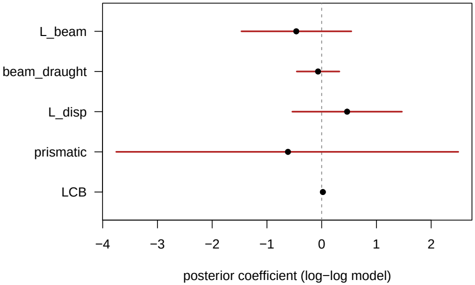
Result: the Froude exponent





- Excellent fit: $R^2 = 0.97$, residual SD $s = 0.32$ on the log scale.
- $\beta_F \approx 4.69$, 95% CI $= [4.60, 4.78]$, $\Pr(\beta_F > 4 \mid \boldsymbol{y}) \approx 1$.
- Power law confirmed — the exponent is *slightly steeper* than the textbook Fn^4.

Hull covariates & residual diagnostics



Hull form factors. Once Fn is in the model, every 95% interval contains zero — the factors vary only across the 22 hulls and are strongly collinear (resolved by BMA in Part 3).

Residual diagnostics expose two defects:

- ∪-shaped trend vs. fitted values — a single constant exponent is too rigid;
- spread shrinks as the fitted value grows — textbook *heteroscedasticity*.

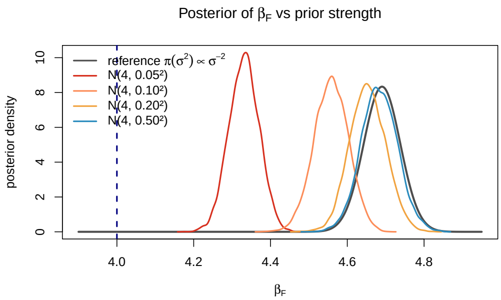
Both motivate the nonlinear correction in Part 4.

Prior sensitivity: does the prior drive $\beta_F > 4$?

Encode the physics in **independent informative priors** and ask how hard they must insist on 4:

$$\beta_F \sim \mathcal{N}(4, \tau^2), \quad \beta_{j \neq F} \sim \mathcal{N}(0, 100^2), \quad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}),$$

sampled with a **Gibbs sampler** (10^4 iters, 2×10^3 burn-in), sweeping τ.



Prior on β_F	**Mean**	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	1.00
reference prior	4.69	1.00

Only a near-dogmatic $\tau = 0.05$ drags the mean to 4.33. For any diffuse prior ($\tau \geq 0.1$) the data dominate: $\beta_F > 4$ is **robust to the prior**.

Part 3

BAS / BMA Model Selection

Model space: which hull factors matter after Froude?

Conditional question. Once the dominant physical predictor is fixed, which hull-geometry variables still carry signal?

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(Fn) + \sum_{k \in \mathcal{M}} \beta_k \, \text{hull}_k + \varepsilon.$$

- $\log(Fn)$ is included in every candidate model.
- Optional hull covariates: `LCB`, `prismatic`, `L_disp`, `beam_draught`, `L_beam`.
- Five optional variables imply $2^5 = 32$ submodels.
- BAS/BMA averages over models instead of relying on one saturated regression.

Question

Among the hull factors, which ones still matter conditional on Froude?

Baseline result: beam-draught has the largest PIP

Posterior inclusion probability (PIP) is the posterior probability that a covariate belongs in the regression model after averaging over all 32 submodels.

Covariate	PIP	Evidence
`beam_draught`	0.565	moderate
`prismatic`	0.265	weak
`LCB`	0.210	weak
`L_disp`	0.206	weak
`L_beam`	0.192	weak

- Highest posterior model: $\log(Fn) + \texttt{beam\_draught}$.
- Its posterior probability is only 0.234, so model uncertainty remains substantial.
- `beam_draught` carries the strongest residual signal, but evidence is not decisive.

Takeaway. Froude is robust; hull effects are weaker, with beam-draught as the best secondary candidate.

Sensitivity: is the BAS/BMA ranking robust to the prior?

The baseline analysis was checked under alternative BAS/BMA prior specifications, with the same 32-model space and $\log(Fn)$ forced into every model.

Prior	LCB	prismatic	L_disp	beam_draught	L_beam
BIC / reference	0.210	0.265	0.206	0.565	0.192
g-prior ($g = n$)	0.186	0.217	0.182	0.499	0.166
JZS	0.060	0.057	0.051	0.243	0.042

- `beam_draught` is top-ranked under all three specifications.
- Its PIP falls from 0.565 to 0.243, so selection is prior-sensitive.
- No hull covariate receives consistently strong support.

Robust conclusion

Not "beam-draught is definitely selected", but "beam-draught is the most defensible secondary hull covariate."

From BAS/BMA to the nonlinear model

$$\text{Log–log regression} \Longrightarrow \text{BAS/BMA over hull factors}$$
$$\Longrightarrow \text{Nonlinear models } M_0 \text{ vs. } M_1.$$

- BAS/BMA proposes `beam_draught` as the best-supported secondary hull covariate.
- Because evidence is moderate, we do not assume it is certainly important.
- Next step:

$$M_0 : \text{Froude only}, \qquad M_1 : \text{Froude} + \texttt{beam\_draught}.$$

- The nonlinear heteroscedastic model checks whether this candidate improves prediction on the original resistance scale.

Takeaway

BAS/BMA is used as a screening step: it narrows the hull variables to the most defensible one, but leaves the final assessment to the nonlinear model comparison.